

Chapter 30 - The x86 Processor

The x86 family began as the Intel 8086 in 1978 and grew into the most widely deployed processor architecture in history. Intuition Engine's x86 processor is 32-bit but not a PC-compatible 386: it implements the complete 8086/8088 instruction set, the 386-era 32-bit register and operand extensions, the x87 floating-point stack, and selected later flat-mode opcodes such as BSWAP and CMOVcc. It runs permanently in a flat real-address mode. Protected mode, paging, segment descriptors, control registers, task-state segments, and the LDT/GDT/IDT machinery are not present.

This makes it a clean target for programmers who already know x86 opcodes but do not need protected mode.

30.1 Register set

30.1.1 General registers

Eight 32-bit general registers, each with 16-bit and 8-bit sub-names that overlap the low bits of the 32-bit form:

32-bit	16-bit	8-bit high	8-bit low	Common use
EAX	AX	AH	AL	Accumulator
EBX	BX	BH	BL	Base register
ECX	CX	CH	CL	Count register
EDX	DX	DH	DL	Data register; high half of multiply/divide
ESI	SI	-	-	Source index
EDI	DI	-	-	Destination index
EBP	BP	-	-	Base pointer (stack frame)
ESP	SP	-	-	Stack pointer

Writes to the 8-bit or 16-bit form leave the unused high bits of the 32-bit register untouched. The four "common use" labels are conventions, not architectural roles, but several instructions do require specific registers (MUL/DIV use EAX/EDX, string ops use ESI/EDI/ECX, etc.).

30.1.2 Segment registers

Six 16-bit segment registers:

Register	Conventional use
CS	Code segment
DS	Default data segment
ES	Extra data segment
SS	Stack segment
FS	Extra data segment (386+)
GS	Extra data segment (386+)

On Intuition Engine all six segments are cleared to 0 on reset, which gives a flat memory model: physical address is just the 32-bit offset. Programs that load a non-zero segment selector get $\text{seg} * 16 + \text{offset}$ (the real-address-mode rule), but the typical case is to leave the segments at zero and address all of memory directly with 32-bit offsets.

30.1.3 Instruction pointer

EIP is the 32-bit instruction pointer. IP is its low 16 bits, used by code that runs in 16-bit mode.

On reset $\text{EIP} = \$00000000$. The historical 8086 reset vector at F000:F000 does not apply in the flat model. Monitor examples in this chapter place code at $\$1000$ and set EIP there explicitly.

Loaded .ie86 flat images use the reset convention. The image bytes are placed starting at physical address $\$00000000$, and execution begins with $\text{EIP} = 0$. If the useful body of a program lives higher in memory, put a small jump or entry stub at address 0 to reach it. That image rule is separate from IE Mon work, where you may set EIP to any address by hand before running.

30.1.4 EFLAGS

The 32-bit EFLAGS register. The low 16 bits are the historical FLAGS register and remain compatible with 8086 and 286 code; the upper bits hold 386-era extensions.

Bit	Mnemonic	Name
0	CF	Carry - status
2	PF	Parity - status
4	AF	Auxiliary carry (BCD) - status
6	ZF	Zero - status
7	SF	Sign - status
8	TF	Trap (single-step) - system
9	IF	Interrupt enable - system
10	DF	Direction (string ops) - control
11	OF	Overflow - status
12-13	IOPL	I/O privilege level - system
14	NT	Nested task - system
16	RF	Resume - system
17	V86	Virtual-8086 mode - system
18	AC	Alignment check - system

Bits 1, 3, 5, and 15 are reserved (bit 1 reads as 1). The IOPL, NT, RF, V86, and AC bits are decoded but have no functional effect because IE has no protected mode.

30.2 Memory model

Address space is 32-bit flat. Multi-byte data is **little-endian**: a 32-bit value $\$12345678$ written to address $\$1000$ places $\$78$ at $\$1000$, $\$56$ at $\$1001$, $\$34$ at $\$1002$, $\$12$ at $\$1003$.

Word and doubleword operands need not be aligned, but unaligned accesses are slower. The MMIO map of Chapter 24 is accessible directly; an `OUT DX, AL` and a `MOV BYTE PTR [F0700], AL` both reach the terminal.

Absolute data operands in `000F0000-000FFFF` reach native MMIO directly. x86 also keeps a smaller compatibility mirror for data accesses in `F000-FFFF`, which maps to `000F0000-000F0FFF`. That mirror is not used for instruction fetch. If `EIP = F000`, the CPU fetches code bytes from flat RAM at `0000F000`, not from the MMIO block.

30.3 Addressing modes

x86 uses the ModR/M and (when scaled-index is needed) SIB byte to encode operand locations. The expressible forms:

Form	Example
Register	<code>MOV EAX, EBX</code>
Immediate	<code>MOV EAX, 12345678</code>
Direct (absolute)	<code>MOV EAX, [F0700]</code>
Register indirect	<code>MOV EAX, [EBX]</code>
Base + displacement	<code>MOV EAX, [EBX+8]</code>
Base + index	<code>MOV EAX, [EBX+ECX]</code>
Base + index*scale	<code>MOV EAX, [EBX+ECX*4]</code>
Base + index*scale + displacement	<code>MOV EAX, [EBX+ECX*4+16]</code>

16-bit addressing forms (`[BX+SI]`, `[BP+DI]`, etc.) are still available with an address-size prefix. 32-bit addressing is the default when no prefix is used.

A segment override prefix (`CS:`, `DS:`, `ES:`, `SS:`, `FS:`, `GS:`) forces a particular segment register. Because all segments resolve to base zero on IE, the override changes which segment register is used in the formula but not the resulting address.

30.4 Instruction set

Appendix G has the full opcode table. The groups below outline what is available.

30.4.1 Data movement

`MOV`, `MOVZX` (zero-extend), `MOVSX` (sign-extend), `CMOVcc` (conditional move to a register), `XCHG`, `BSWAP` (32-bit byte swap), `XLAT` (`AL = [EBX + AL]`), `PUSH/POP`, `PUSHA/POPA`, `PUSHF/POPF`, `LEA`, `LDS/LFS/LGS/LSS` (load far pointer).

`CMOVcc` uses the same condition names as `Jcc`: `CMOVZ`, `CMOVNZ`, `CMOVB`, `CMOVNB`, `CMOVL`, `CMOVNL`, and the other flag tests. The destination is always a register and the source may be a register or memory. A memory source is read even when the condition is false, so do not use an untaken `CMOVcc` to avoid reading a read-sensitive MMIO register.

30.4.2 Arithmetic

`ADD`, `ADC`, `SUB`, `SBB`, `INC`, `DEC`, `NEG`, `CMP`. Multiply: `MUL` (unsigned), `IMUL` (signed, three forms). Divide: `DIV` (unsigned), `IDIV` (signed). Decimal: `DAA`, `DAS`, `AAA`, `AAS`, `AAM`, `AAD`. Sign/zero extend `AL` to `AX`, `AX` to `EAX`: `CBW/CWDE`. Sign-extend `EAX` to `EDX:EAX`: `CDQ`.

30.4.3 Logic, shifts, rotates

AND, OR, XOR, NOT, TEST. Shifts: SHL, SHR, SAL, SAR, SHLD, SHRD (386+ double-precision shifts). Rotates: ROL, ROR, RCL, RCR.

30.4.4 Bit and bit-string

BT, BTS, BTR, BTC: bit test / test-and-set/reset/complement. BSF, BSR: bit scan forward / reverse. SETcc: set byte on condition.

30.4.5 String

MOVS, CMPS, SCAS, LODS, STOS, INS, OUTS. Each has explicit (MOVSB/MOVSW/MOVSDB) and implicit forms. The REP, REPE, and REPNE prefixes repeat using ECX as a counter.

30.4.6 Control transfer

Mnemonic	Effect
JMP	Unconditional jump (relative, register-indirect, memory-indirect, far)
Jcc	Conditional jump (JE/JZ, JNE/JNZ, JL, JG, JC, JNC, JO, JNO, JS, JNS, JP, JNP, JA, JAE, JB, JBE, JCXZ, JECXZ)
LOOP	Decrement ECX, jump if non-zero
LOOPE/LOOPNE	Decrement and conditionally jump
CALL	Call subroutine (near or far)
RET	Return
INT n	Software interrupt (n = 0..255)
INT0	Trap if 0F set
IRET/IRETD	Return from interrupt

30.4.7 Port I/O

Mnemonic	Operation
IN AL, imm8	Read 8-bit port imm8 into AL
IN AX, imm8	Read 16-bit port
IN EAX, imm8	Read 32-bit port
IN AL, DX	Read port DX (allows full 16-bit port number)
OUT imm8, AL	Write AL to port imm8
OUT DX, AL	Write AL to port DX
INS/OUTS	Block port I/O (with REP prefix)

Intuition Engine assigns ports per device; the device chapters list the relevant port numbers. Many devices are also reachable through memory-mapped registers and most x86 code uses those directly.

The useful x86 port assignments are:

Ports	Device
\$60-\$69	POKEY direct register window
\$A0-\$AD	VGA registers
\$B0-\$B7	Voodoo register and texture gateway
\$D4/\$D5	ANTIC select/data
\$D6/\$D7	GTIA select/data
\$E0/\$E1	SID select/data
\$F0/\$F1	PSG select/data
\$F2/\$F3	TED select/data
\$FE	ULA border port

30.4.8 Flag manipulation

CLC, STC, CMC (clear/set/complement carry); CLI, STI (disable/enable interrupts); CLD, STD (clear/set direction); LAHF, SAHF (load/store AH from/to flags); PUSHF/POPF and their 32-bit PUSHFD/POPFD siblings.

30.4.9 Other

NOP, HLT, WAIT, CPUID (returns vendor and feature bits).

The x87 floating-point coprocessor is present; FPU opcodes D8-DF execute against an 80-bit stack of eight registers ST(0)-ST(7). The environment operations used to save and restore that state are described below and their hand-entry forms are listed in Appendix G.

30.4.10 x87 environment operations

The x87 environment is a little-endian block. FLDENV reads the first 28 bytes and FNSTENV writes them using this layout:

Offset	Size	Field
0	4	Control word in the low 16 bits.
4	4	Status word in the low 16 bits.
8	4	Tag word in the low 16 bits.
12	4	Instruction pointer.
16	4	Code selector in the low 16 bits and opcode in bits 16-26.
20	4	Data pointer.
24	4	Data selector in the low 16 bits.

The tag word preserves the individual register classifications, including empty, zero, and special values. FNSTENV writes the environment before setting all six exception-mask bits in the live control word. FLDENV restores the fields from memory.

FNSAVE uses the same 28-byte environment at the start of its operand, then writes eight 80-bit register values at offsets 28, 38, 48, 58, 68, 78, 88, and 98. Its complete save size is 108 bytes. FRSTOR reads the same 108-byte layout. FNSAVE resets the x87 after the save, so a programme which needs the values again must use FRSTOR.

The memory forms use the normal x86 ModR/M byte. The /4 form selects FLDENV or FRSTOR, and the /6 form selects FNSTENV or FNSAVE. For an absolute 32-bit address, use ModR/M \$25 for /4 or \$35 for /6, followed by the address in little-endian order. For example, D9 35 00 50 00 00 is FNSTENV [\$00005000], and DD 35 00 50 00 00 is FNSAVE [\$00005000].

30.5 Interrupts

The interrupt vector table sits at physical address \$00000000, just like the 8086 real-mode IVT. Each entry is four bytes: a 16-bit IP followed by a 16-bit CS.

When INT *n* executes or a hardware interrupt fires:

1. The CPU pushes the low 16 bits of EFLAGS, then CS, then IP.
2. Clears IF and TF.
3. Loads IP from [4*n] and CS from [4*n + 2].

IRET reverses the push order, popping IP, CS, and FLAGS.

NMI is vector 2. Interrupt vectors 0-7 and 16-19 are reserved by the architecture (divide error, debug, NMI, breakpoint, overflow, bounds, invalid opcode, device-not-available, and FPU diagnostics).

30.6 What is not implemented

By design IE's x86 omits:

- Protected mode and the descriptor tables (GDT, IDT, LDT, TSS).
- Paging and the CR0-CR4 control registers.
- Real-mode BIOS interrupts. The IVT exists but contains zero vectors at reset; a program is free to install its own.
- The historical reset vector at F000:FFF0. IE starts at EIP = 0; monitor-entered examples normally set EIP to their chosen start address.
- SMM, VMX, all post-486 MSR machinery.

Programs that need any of these would have to be written for a different processor. For ordinary applications, the flat 32-bit real-address model is the most pleasant x86 environment available.

30.7 A small example

This x86 byte-entry program uses byte stores to make TED play two detuned square voices. In flat IE x86 mode, MOV [addr], AL reaches the MMIO address directly:

```

(x86)> w 1000 B0 01 A2 00 08 0F 00 B0 1C A2 00 0F 0F 00 B0 02
(x86)> w 1010 A2 04 0F 0F 00 B0 58 A2 01 0F 0F 00 B0 02 A2 02
(x86)> w 1020 0F 0F 00 B0 38 A2 03 0F 0F 00 EB FE
(x86)> d 1000 #13
001000: B0 01          MOV AL, $01
001002: A2 00 08 0F 00  MOV [$000F0800], AL
001007: B0 1C          MOV AL, $1C
001009: A2 00 0F 0F 00  MOV [$000F0F00], AL
00100E: B0 02          MOV AL, $02
001010: A2 04 0F 0F 00  MOV [$000F0F04], AL
001015: B0 58          MOV AL, $58
001017: A2 01 0F 0F 00  MOV [$000F0F01], AL
00101C: B0 02          MOV AL, $02
00101E: A2 02 0F 0F 00  MOV [$000F0F02], AL
001023: B0 38          MOV AL, $38
001025: A2 03 0F 0F 00  MOV [$000F0F03], AL
T 00102A: EB FE          JMP SHORT $0000102A
(x86)> r eip 1000
(x86)> b 102A
(x86)> g
(x86)> m F0F00 1
000F0F00: 1C 58 02 38 02 00 00 00 00 00 00 00 00 00 00 00 .X.8.....
(x86)> bc 102A

```

The byte stream is small enough to type and inspect directly:

Address	Bytes	Meaning
\$1000	B0 01	MOV AL, 1; opcode \$B0 loads an immediate byte into AL.
\$1002	A2 00 08 0F 00	MOV [\$000F0800], AL; opcode \$A2 stores AL at an absolute little-endian address. This enables audio.
\$1007	B0 1C	Loads TED voice 1 low divider byte \$1C.
\$1009	A2 00 0F 0F 00	Writes to \$F0F00, TED_FREQ1_LO.
\$100E	B0 02	Loads TED voice 1 high divider bits \$02.
\$1010	A2 04 0F 0F 00	Writes to \$F0F04, TED_FREQ1_HI.
\$1015	B0 58	Loads TED voice 2 low divider byte \$58.
\$1017	A2 01 0F 0F 00	Writes to \$F0F01, TED_FREQ2_LO.
\$101C	B0 02	Loads TED voice 2 high divider bits \$02.
\$101E	A2 02 0F 0F 00	Writes to \$F0F02, TED_FREQ2_HI.
\$1023	B0 38	Loads TED sound control \$38, both voices on with volume 8.
\$1025	A2 03 0F 0F 00	Writes to \$F0F03, TED_SND_CTRL.
\$102A	EB FE	Short jump back by 2, which branches to \$102A and keeps the sound active.

The first store enables audio. TED voice 1 gets divider \$21C through low byte \$1C and high bits \$02; voice 2 gets divider \$258 through low byte \$58 and high bits \$02. \$38 in TED_SND_CTRL enables both voices and sets volume 8, the chip's maximum. The address fields after each A2 opcode are little-endian: \$000F0F03 is entered as 03 0F 0F 00.

30.8 TED video example

The same two x86 instruction forms can draw on the TED video chip. This program enables TED video, sets the border and background colours, puts character code 1 in the top-left matrix cell, assigns that cell a foreground colour, and replaces character 1 with an 8-byte motif.


```

(x86)> w 1100 B0 01 A2 58 0F 0F 00 B0 18 A2 20 0F 0F 00
(x86)> w 110E B0 08 A2 24 0F 0F 00 B0 06 A2 30 0F 0F 00
(x86)> w 111C B0 2E A2 40 0F 0F 00 B0 01 A2 00 30 0F 00
(x86)> w 112A B0 4E A2 00 34 0F 00 B0 FF A2 08 38 0F 00
(x86)> w 1138 B0 81 A2 09 38 0F 00 B0 BD A2 0A 38 0F 00
(x86)> w 1146 B0 A5 A2 0B 38 0F 00 B0 A5 A2 0C 38 0F 00
(x86)> w 1154 B0 BD A2 0D 38 0F 00 B0 81 A2 0E 38 0F 00
(x86)> w 1162 B0 FF A2 0F 38 0F 00 EB FE
(x86)> d 1100 #31
001100: B0 01          MOV AL, $01
001102: A2 58 0F 0F 00 MOV [$000F0F58], AL
001107: B0 18          MOV AL, $18
001109: A2 20 0F 0F 00 MOV [$000F0F20], AL
00110E: B0 08          MOV AL, $08
001110: A2 24 0F 0F 00 MOV [$000F0F24], AL
001115: B0 06          MOV AL, $06
001117: A2 30 0F 0F 00 MOV [$000F0F30], AL
00111C: B0 2E          MOV AL, $2E
00111E: A2 40 0F 0F 00 MOV [$000F0F40], AL
001123: B0 01          MOV AL, $01
001125: A2 00 30 0F 00 MOV [$000F3000], AL
00112A: B0 4E          MOV AL, $4E
00112C: A2 00 34 0F 00 MOV [$000F3400], AL
001131: B0 FF          MOV AL, $FF
001133: A2 08 38 0F 00 MOV [$000F3808], AL
001138: B0 81          MOV AL, $81
00113A: A2 09 38 0F 00 MOV [$000F3809], AL
00113F: B0 BD          MOV AL, $BD
001141: A2 0A 38 0F 00 MOV [$000F380A], AL
001146: B0 A5          MOV AL, $A5
001148: A2 0B 38 0F 00 MOV [$000F380B], AL
00114D: B0 A5          MOV AL, $A5
00114F: A2 0C 38 0F 00 MOV [$000F380C], AL
001154: B0 BD          MOV AL, $BD
001156: A2 0D 38 0F 00 MOV [$000F380D], AL
00115B: B0 81          MOV AL, $81
00115D: A2 0E 38 0F 00 MOV [$000F380E], AL
001162: B0 FF          MOV AL, $FF
001164: A2 0F 38 0F 00 MOV [$000F380F], AL
T 001169: EB FE          JMP SHORT $00001169
(x86)> r eip 1100
(x86)> b 1169
(x86)> g
(x86)> m F0F20 1
000F0F20: 18 00 00 00 00 08 00 00 00 20 00 00 00 00 00 00 .....
(x86)> m F0F24 1
000F0F24: 08 00 00 00 00 20 00 00 00 00 00 00 06 00 00 00 ....
(x86)> m F3000 1
000F3000: 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
(x86)> m F3400 1
000F3400: 4E 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 N.....
(x86)> m F3808 1
000F3808: FF 81 BD A5 A5 BD 81 FF 00 00 00 00 00 00 00 00 .....
(x86)> bc 1169

```

The first five stores configure TED video:

Address	Value	Meaning
\$F0F58	\$01	Enable TED video output.
\$F0F20	\$18	DEN + RSEL: display enabled, 25-row character field.
\$F0F24	\$08	CSEL: 40-column character field.
\$F0F30	\$06	Background colour 0.
\$F0F40	\$2E	Border colour.

The next two stores set up the top-left cell. \$F3000 is the first byte of the video matrix, and \$F3400 is the matching colour RAM byte. The eight stores at \$F3808-\$F380F replace the glyph for character 1; the bytes form a small block motif.

After the breakpoint is reached, the top-left TED character cell shows that motif in colour \$4E on background colour \$06, with the border set to \$2E.

30.9 What comes next

Chapter 31 returns to material that applies across every CPU: how each processor handles timing, traps, and exceptions; how the shared timer counter is exposed to each one; and how the IRQ priority levels intersect with the auto-vector and MMU exception paths.